

Ex. Nos. 4	Demonstration of Hash Cracking Using John the Ripper Tool in Kali Linux
Date :	

Aim

To demonstrate password hash cracking using the John the Ripper tool in Kali Linux for understanding password vulnerabilities and security analysis.

Software Requirements

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Security Tool: John the Ripper
- System Requirements:
 - o RAM: Minimum 4 GB (8 GB recommended)
 - o Disk Space: Minimum 30 GB
 - o Processor: Intel/AMD with Virtualization support

Introduction

Password hashes are commonly used to store user credentials securely. However, weak passwords and outdated hashing techniques can be exploited using password-cracking tools. John the Ripper is a popular open-source password auditing and cracking tool used to identify weak passwords by performing dictionary, brute-force, and hybrid attacks. Using Kali Linux in a virtualized environment allows safe and ethical experimentation without affecting real systems.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Open Terminal

1. Click on Terminal from the Kali Linux menu.
2. Ensure John the Ripper is installed:
3. `john --version`

Step 3: Prepare the Hash File

1. Create a text file containing password hashes (example: hash.txt).
2. Store the hash inside the file:(get some hashcode from net and type in file)
3. `nano hash.txt`
4. Save and exit the file.

Step 4: Run John the Ripper

1. Execute the John the Ripper command:
2. `john hash.txt`
3. The tool starts cracking the hash using default wordlists.

Step 5: View Cracked Passwords

1. To display cracked passwords:
2. `john --show hash.txt`

Output

- John the Ripper successfully cracks weak password hashes.
- The cracked password is displayed in plaintext format.

```
1000 packages can be upgraded... run 'apt list --upgradable' to see them.

(student@kali)~[~/NearNow]
$ nano hash.txt
e99a18c428cb38d5f260853678922e03

(student@kali)~[~/NearNow]
$ john hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (MD5 [32/64])
Will run 4 OpenMP threads
Press 'q' or Ctrl+C to abort, almost any other key for status
rec123 (?)
1g 0:00:00:02 DONE (2026-04-18 16:45) 0.5000g/s 1200p/s 1200c/s 1200C/s rec13 .122456
Use the "--show" option to display all of the cracked passwords reliably
Session completed

(student@kali)~[~/NearNow]
$ john --show hash.txt
?:rec123
1 password hash cracked, 0 left

(student@kali)~[~/NearNow]
$
```

Result

Hash cracking was successfully demonstrated using the John the Ripper tool in Kali Linux. The experiment highlights the importance of using strong passwords and secure hashing algorithms.